

Demonstration Plan - Canny Edge Detection

Live Webcam (Accelerated) Demo

First we will demonstrate our FPGA-accelerated Canny edge detector running on the live input of a webcam. Both what the webcam sees and what the output of our algorithm will be displayed in real-time. This also demonstrates how our solution can easily be added to existing computer vision pipelines using OpenCV.

OpenCV Benchmark

Then we will benchmark OpenCV's builtin Canny edge detection algorithm. This is a multithreaded algorithm that runs on the 4 ARM cores on the KV260. To ensure a fair comparison, we ensure that the CPU frequency scaling factor is set to be the maximum. The benchmark loads a static 640x480 image, and (after a warmup iteration) runs 1000 timed iterations of the algorithm, and then prints the overall statistics.

FPGA Accelerated Benchmark

Then we will compare the results from the OpenCV benchmark to the benchmark of our FPGA-accelerated algorithm. The test runs in the exact same sequence as the OpenCV benchmark, ensuring fairness.

OpenCV-FPGA Difference

Finally, we will compare the edges detected by our accelerated algorithm compared to OpenCV. Since our algorithm runs on fixed hardware, it cannot run the exact same hysteresis algorithm (flood-fill), so we expect some difference along weak edges that our algorithm drops.